The Periodic Constantin–Lax–Majda Equation: Exact Arbitrary-Mean Riccati Flow

HCS-C377 theorem package

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Abstract

We solve the inviscid nonadvective Constantin–Lax–Majda equation on the circle for arbitrary conserved mean under one frozen periodic Hilbert transform convention. The periodic Tricomi identity turns the nonlocal PDE into a pointwise complex Riccati equation, while an invariant Hardy subspace proves that the Hilbert constraint is preserved. This yields exact Möbius formulas in both the zero- and nonzero-mean strata.

Keywords: periodic Hilbert transform; Constantin–Lax–Majda equation; complex Riccati flow; exact Möbius solution; Hardy constraint; conserved mean.

中文摘要

本文在固定的周期希尔伯特变换符号约定下，求解任意守恒均值的无粘、 无平流康斯坦丁、拉克斯和马吉达方程。周期特里科米恒等式把非局部方程 化为逐点复黎卡提方程，而不变哈代子空间保证希尔伯特约束始终保持， 由此得到零均值与非零均值两类精确莫比乌斯公式。

关键词：周期希尔伯特变换；康斯坦丁、拉克斯和马吉达方程； 复黎卡提流；精确莫比乌斯解；哈代约束；守恒均值。

1 Convention and exact Riccati closure

Write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. For a real smooth periodic function, define

$$\mathcal{H}(\mathrm{e}^{\mathrm{i}kx}) = -\mathrm{i}\,\mathrm{sgn}(k)\mathrm{e}^{\mathrm{i}kx}, \qquad \mathcal{H}1 = 0. \tag{1}$$

Thus $\mathcal{H}\sin(kx) = -\cos(kx)$ and $\mathcal{H}\cos(kx) = \sin(kx)$ for $k > 0$. On mean-zero functions, $\mathcal{H}^2 = -I$; on real L^2, $\mathcal{H}$ is skew-adjoint.

Consider

$$\partial_t\omega = \omega\mathcal{H}\omega, \qquad \omega(0,x) = \omega_0(x). \tag{2}$$

Let

$$\mu = \langle\omega_0\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi}\omega_0(x)\,\mathrm{d}x, \quad f = \omega - \mu, \quad h = \mathcal{H}f, \quad z = h + \mathrm{i}f.$$

Lemma 1 (Periodic Tricomi identity with the fixed sign)**.** *For real mean-zero f,*

$$\mathcal{H}(f\mathcal{H}f) = \frac{(\mathcal{H}f)^2 - f^2}{2}. \tag{3}$$

Moreover $\langle f\mathcal{H}f\rangle = 0$.

Proof. For a trigonometric polynomial, split its Fourier series into positive and negative Hardy parts. On those parts (1) is multiplication by $-\mathrm{i}$ and $+\mathrm{i}$, respectively. Comparing the positive and negative parts of the product gives (3); the zero Fourier coefficient vanishes by the skew-adjointness of $\mathcal{H}$. Density and boundedness of $\mathcal{H}$ on Sobolev spaces extend the identity to smooth f. □

Theorem 1 (Arbitrary-mean pointwise flow)**.** *The mean μ is conserved and*

$$z_t = \mathrm{i}\mu z + \frac{1}{2}z^2. \tag{4}$$

Before the first zero of its denominator, the exact solution is

$$\mu = 0: \quad z(t,x) = \frac{2z_0(x)}{2 - tz_0(x)}, \tag{5}$$

$$\omega(t,x) = \frac{4\omega_0(x)}{[2 - th_0(x)]^2 + t^2\omega_0(x)^2}; \tag{6}$$

$$\mu \neq 0: \quad z(t,x) = \frac{\mathrm{e}^{\mathrm{i}\mu t} z_0(x)}{1 - [\mathrm{e}^{\mathrm{i}\mu t} - 1] z_0(x)/(2\mathrm{i}\mu)}. \tag{7}$$

If

$$\Delta(t,x) = 2\mathrm{i}\mu - [\mathrm{e}^{\mathrm{i}\mu t} - 1] z_0(x), \tag{8}$$

then the real vorticity formula for $\mu \neq 0$ is

$$\omega(t,x) = \frac{4\mu^2\omega_0(x)}{|\Delta(t,x)|^2}. \tag{9}$$

Proof. Skew-adjointness gives $\partial_t\langle\omega\rangle = \langle\omega\mathcal{H}\omega\rangle = 0$. Hence

$$f_t = \mu h + fh, \qquad h_t = \mu\mathcal{H}h + \mathcal{H}(fh) = -\mu f + \tfrac{1}{2}(h^2 - f^2),$$

where Lemma 1 and $\mathcal{H}h = -f$ were used. Combining the two real equations proves (4). Separation gives (5). For $\mu \neq 0$, differentiating $1/z$ gives a scalar linear equation; solving it yields (7). Taking the imaginary part of (5) gives (6). Finally multiply (7) by $2\mathrm{i}\mu/\Delta$ and its complex conjugate. Using $z_0 = h_0 + \mathrm{i}(\omega_0 - \mu)$ gives exactly (9).

It remains to check that the pointwise Riccati solution is still a solution of the nonlocal PDE, rather than merely of a larger product system. For $s > 1/2$, the negative-frequency Hardy subspace of $H^s(\mathbb{T})$ is a Banach algebra. Under (1), $z_0 = \mathcal{H}f_0 + \mathrm{i}f_0$ has only strictly negative Fourier modes. The vector field $z \mapsto \mathrm{i}\mu z + z^2/2$ is locally Lipschitz and preserves that subspace, so uniqueness for the Banach-space ODE shows that the explicit $z(t)$ retains strictly negative modes. Consequently its imaginary part has mean zero and its real part is exactly the Hilbert transform of its imaginary part: $h = \mathcal{H}f$.

Equivalently, if $g = h - \mathcal{H}f$ is the constraint defect in the product system, Lemma 1 gives

$$g_t = (\mathcal{H}f)g + \tfrac{1}{2}g^2 - \mu\mathcal{H}g - \mathcal{H}(fg).$$

This equation is locally Lipschitz in the same Sobolev algebra, and $g(0) = 0$ forces $g \equiv 0$. On any compact time interval where the displayed denominator has a positive lower bound, the explicit rational formula remains smooth in x; the same local uniqueness argument continues the PDE solution across the interval. Thus a denominator zero, not a loss of the Hilbert constraint, is the only obstruction used below. □

Round-zero certificate: round zero fixes the periodic Hilbert sign, proves the Tricomi closure, conserves the arbitrary mean, and derives both exact Möbius solutions and real vorticity formulas.